

Cross-View Identity in Image-Based Fruit Counting: A Systematic Review and Design Space for Class-Wise Inventories

Muhammad Zainal Muttaqin and Fatma Indriani

Abstract—Image-based fruit inventories count fruit from photographs, but a camera moving along a row or around a tree sees most fruit several times, so a sum of detections overstates the true number. This review examines how the literature decides which observations belong to the same fruit and how class labels such as maturity are attached to each unique fruit. A systematic search of Scopus identified 971 relevant studies published between 2012 and 2026. The 187 studies that combine several observations of the same fruit were classified into six association mechanisms: no association, statistical correction, appearance matching, temporal tracking, geometric or 3D association, and learned association. Temporal tracking dominates recent work, and geometric association provides the most direct calibration-free counts when camera pose and depth are reliable. On fruit that look alike, appearance features did not improve tracking over motion alone. Few studies report identity errors or counts per class, and comparisons of mechanisms on the same data are rare. In oil palm, published work concentrates on ripeness classification and bunch detection in single images; cross-view association of bunches on a palm has not been evaluated. The review relates each mechanism to the acquisition conditions it requires and proposes the measurements needed to evaluate duplicate-aware, class-wise inventories.

Index Terms—fruit counting, multi-view perception, multi-object tracking, data association, structure from motion, yield estimation, oil palm, systematic review

I. INTRODUCTION

A grower who needs the number of ripe fruit on a block wants one record for each physical fruit, with its class. A camera does not supply that record. It supplies images, and in most field protocols the same fruit appears in several of them: in consecutive video frames, in photographs taken from both sides of a row, or in images taken while walking around a single tree. Detection finds fruit in each image. It does not say whether a fruit found in one image is the fruit found in another. The step that makes that decision determines whether a sum of detections is an inventory or an overcount.

The size of the problem is known from studies that measured it directly. In a mango orchard, one image of each side of 16 trees counted 54% of the fruit that were hand-counted, and one image of one side counted 27%; tracking every fruit through a multi-view sequence and triangulating it counted 101.4% [1]. On 21 mango trees, video tracking recorded 62% of the harvest count and two images per tree recorded 40% [2]. On a 192-fruit video, the same tracker produced 9.9% double counts and 7.3% missed fruit, which netted to a 2.6%

overcount [2]. The last figure shows why a total count can look correct when the underlying identity decisions are not: a duplicate and a miss cancel.

In oil palm the problem takes a specific form. Plantations estimate output with a bunch census in which staff count and classify bunches on each palm by the number of months until harvest. The SawitMVC dataset photographed 953 palms from four or eight sides and annotated 9,823 unique bunches in four census classes, with links that record which boxes in different images show the same bunch [3]. A census built from such photographs needs both the identity decision and a class label for each unique bunch.

Earlier reviews organize this literature by sensor [4], [5], by detector family [6], [7], by yield-estimation route [8], [9], or, for oil palm, by ripeness-sensing method [10], [11]. None of them, as far as their stated scope shows, takes the identity of a fruit across observations as the organizing variable. This review does. It asks four questions:

- RQ1 Which mechanisms have been used to decide that observations in different images belong to the same fruit, and what acquisition conditions does each assume?
- RQ2 What evidence shows how well each mechanism controls duplicates and misses, and how was that evidence measured?
- RQ3 How are class attributes, mainly maturity, attached to a unique fruit rather than to each detection?
- RQ4 What does the oil-palm imaging literature cover, and what is missing for a bunch-level, class-wise census from images?

The contributions are threefold. First, a systematic map of 971 Scopus-indexed studies from 2012 to 2026, with the search strings reported in full and the screening decisions and codes provided as supplementary material (Section II). Second, a design space that separates acquisition, observation, association, attribute assignment, and inventory, and classifies association into six mechanisms, each tied to the conditions it assumes (Section III). Third, a synthesis of the 187 multi-observation studies by mechanism, with the comparisons that were made on the same data reported in two tables, followed by the evidence on class attributes, depth, and oil palm (Sections IV–VIII). Table IV places the review against earlier reviews, and Section X lists the measurements that would let a future study claim a duplicate-aware, class-wise inventory.

TABLE I
SCOPUS QUERY SETS. FULL STRINGS ARE GIVEN IN APPENDIX B.

Run	Target	Hits	Kept
Q1	Fruit or crop terms AND counting or yield AND multi-view, video, tracking, SfM, SLAM, duplicates	871	871
Q2	Fruit AND counting or load in the title	757	757
Q3	Oil palm or FFB AND imaging, learning, ripeness, or counting	1,079	1,079
Q4	Fruit AND ripeness, grade, or class AND counting AND detection	361	361
Q5	Fruit AND RGB-D, stereo, LiDAR, or point cloud AND detection or counting	1,988	1,988
Q6	Reviews of MOT, re-ID, multi-view detection, SfM, SLAM	236	236
Q7	Reviews of fruit detection, counting, ripeness, yield	1,049	1,049
Q8a-e	MOT, multi-view, SfM/SLAM, object counting, re-ID (titles)	23,532	125
Q9	Named foundational methods (SORT, NeRF, SAM, ...)	1,082	25
Total		6,491	

II. REVIEW METHOD

A. Review type and reporting

The review is a systematic mapping review with a configurative synthesis [12], [13]. It maps a broad body of work and builds a classification from it; it does not pool effect sizes, because the included studies use different crops, references, and metrics. Reporting follows PRISMA 2020 where the items apply to a mapping review [14], [15], and the protocol elements recommended for literature reviews in engineering and management were used to decide what to report [16], [17].

B. Source and search

Scopus was used as the single bibliographic database because it indexes the journals and proceedings in which agricultural engineering and computer vision work is published, including those of Elsevier, IEEE, Springer, MDPI, and Frontiers. The search was run on 28 September 2026 (UTC). The search comprised the query sets summarised in Table I. Q1–Q5 target agricultural fruit perception: multi-observation counting, counting in titles, oil-palm imaging, class attributes with counting, and depth or 3D sensing. Q6 and Q7 retrieve reviews: methods reviews on tracking, re-identification, multi-view detection, and reconstruction, and agricultural reviews on fruit detection and yield. Q8a–Q8e and Q9 retrieve methods papers outside agriculture; they were sorted by citation count and only the 25 most cited records of each run were retained, because their role is to identify the transferable mechanisms, not to map those fields. All runs were limited to 2012–2026 and to subject areas that publish engineering or agricultural work. The complete search strings are given in Appendix B.

C. Eligibility criteria

A record was included if it reported an image-based method for fruit on the plant, or on the ground under the plant in the case of oil-palm loose fruit, and evaluated counting,

yield estimation, detection with a class attribute, localization, or sizing; or if it was a review of these topics; or if it was a methods paper on tracking, re-identification, multi-view association, 3D reconstruction, or object counting outside agriculture. Records were excluded if the object was not fruit on the plant (for example trees, leaves, or flowers only); if imaging was postharvest or laboratory-only, except oil-palm mill grading, which is part of the case; if yield was modelled without fruit-level detection; if the work addressed only picking or manipulation; if fruit were detected without counting or any evaluation beyond detection; if the sensor was not an imaging sensor; or if the paper was not in English.

D. Screening

Duplicates were removed by Scopus EID. Proceedings front matter, errata, notes, editorials, letters, and retracted items were removed by document type. Titles were then screened; for the three largest runs (Q3, Q5, Q7) titles were first sorted by keyword so that similar titles were read together. Abstracts were compiled from OpenAlex, Semantic Scholar, Crossref, and Europe PMC. For 254 records no abstract could be found; these were assessed from the title, the source, and, where available, the summary provided by Semantic Scholar. Each included study was assigned to one of seven groups: multi-observation studies, which combine several observations of the same fruit (video, several views, or repeated scans); single-view counting and yield studies; oil-palm fresh-fruit-bunch (FFB) imaging studies, including mill grading and loose fruit; studies that count fruit by class in single images; depth, 3D, and non-RGB sensing studies; earlier reviews; and methods papers from outside agriculture. Screening was carried out by one reviewer working with a large language model, which read the titles and abstracts and recorded a decision and a code for each record. No second, independent reviewer checked the decisions; this limitation is discussed in Section XI.

E. Data extraction and coding

All 187 multi-observation studies were coded individually from the abstract and, where open-access full text was obtained, from the paper: identity mechanism (Section III), acquisition (video along a path, discrete views, 3D scan, revisits over time, or a single view), whether counts are reported per class, and the main reported result. The 170 oil-palm studies were coded individually for task, setting, modality, use of several views, and whether class labels are attached to instances. The remaining groups were coded by keyword rules for crop, modality, platform, and class attribute, and the rule output was spot-checked. Metric types were extracted from abstracts by keyword and are reported only for studies with an abstract. Full text was obtained for the 331 included studies available in open access.

F. Appraisal of evidence

No formal quality score was assigned. Instead, the review records, for each comparison it reports, the reference against which counts were checked (hand count on the tree, harvest

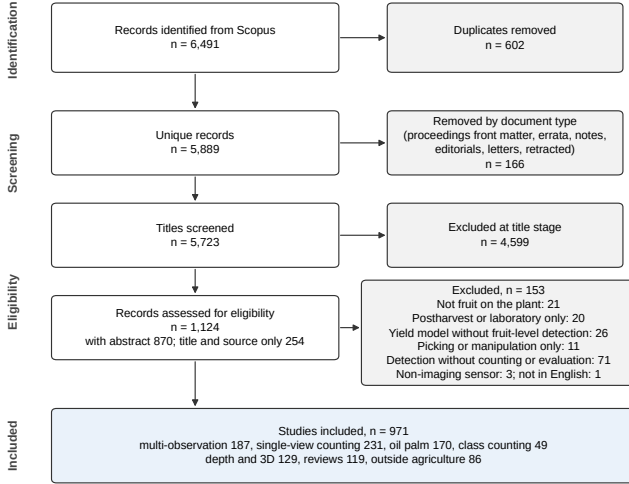


Fig. 1. Flow of records through the review (PRISMA 2020).

count, packhouse count, or annotated images), whether the comparison was made on the same trees or data, and whether the result is a total count, a per-tree count, or an identity metric. These three properties decide what a reported number can support.

G. Result of the search

Figure 1 gives the flow. The searches returned 6,491 records, of which 5,889 were unique. After 166 were removed by document type, 5,723 titles were screened and 4,599 excluded. Of the 1,124 records assessed for eligibility, 153 were excluded, most often (71 records) because they detected fruit without counting or evaluating anything beyond detection. The map contains 971 studies. Figure 2 shows their growth, from 8 records in 2012 to 185 in 2025.

III. A DESIGN SPACE FOR TURNING OBSERVATIONS INTO RECORDS

A. From images to an inventory

The pipeline that ends in a class-wise inventory has five stages (Figure 3). Acquisition fixes how many views of each plant exist, whether they are ordered in time, and whether pose, depth, or baseline are known. Observation produces detections or masks in each image; a fruit may produce several observations, and a detector may also split one fruit or merge two. Association decides which observations belong to the same physical fruit. Attribute assignment gives the unique fruit a class, size, or mass. The inventory aggregates unique fruit by class and by tree, row, or block. Detection accuracy, which most papers report, describes the second stage. The count, which the user needs, depends on all five.

B. Three sources of double counting

Liu *et al.* named three ways in which a fruit is counted twice in a traversal: the same fruit detected in consecutive images, the same fruit tracked, lost, and picked up again as a new

track, and the same fruit seen from the two sides of a tree [18]. Wang *et al.* cited the same three sources when designing their mango tracker [2]. Misses have two sources: fruit never visible from any camera position, and fruit visible but not detected. A mechanism controls some of these sources and not others, which is why the mechanisms below are described by what they assume, not only by how they compute.

C. Six association mechanisms

M0, no association. Each image is counted separately. Correct only if acquisition guarantees that no fruit appears twice, or if duplicates are accepted.

M1, statistical correction. Image counts are summed or averaged and then related to hand or harvest counts by a fitted factor or regression. No individual fruit is identified. It assumes that the ratio between observed and true counts is stable across the trees to which it is applied [19], [20].

M2, appearance matching. Observations are linked when their appearance descriptors are similar, either hand-crafted or learned for re-identification. It assumes that fruit differ enough in appearance to be told apart.

M3, temporal tracking. Observations in consecutive frames are linked by predicted motion and overlap, usually with a Kalman filter and an assignment step [21]; a counting line or region then decides when a track is counted. It assumes that consecutive frames overlap and that motion between them is small and smooth.

M4, geometric or 3D association. Observations are linked through a common coordinate frame: triangulation with known camera poses, structure from motion, SLAM, registered point clouds, or an implicit 3D representation such as a neural radiance field. It assumes that poses, scale, and depth or reconstruction are accurate enough that two different fruit are not placed at the same position.

M5, learned association. A network is trained to decide correspondence directly from linked examples. It assumes that training data with identity links exist and represent the conditions of use.

A study may combine mechanisms; 55 of the 182 multi-observation method studies do (Table V). A study that combines them should still show which one removes the duplicates, because each fails under different conditions.

D. What the 187 multi-observation studies use

Of the 187 multi-observation studies, five are dataset papers and 182 propose or evaluate a method. Temporal tracking (M3) appears in 113 of the 182 (62%), geometric association (M4) in 67 (37%), appearance matching (M2) in 32 (18%), statistical correction (M1) in 18 (10%), no association (M0) in 5, and learned association (M5) in 3. Most studies acquire video along a path (135), 37 use discrete views, five use 3D scans, and three revisit the same plants over time. The mix has shifted: before 2018, M1 and M4 were the only mechanisms in use; since 2023, tracking appears in about 70% of new studies (Figure 4a). Apple accounts for 53 studies, citrus and grape for 20 each, tomato for 15, strawberry for 10, and mango for 7 (Figure 5). Only 19 of the 182 report counts per class.

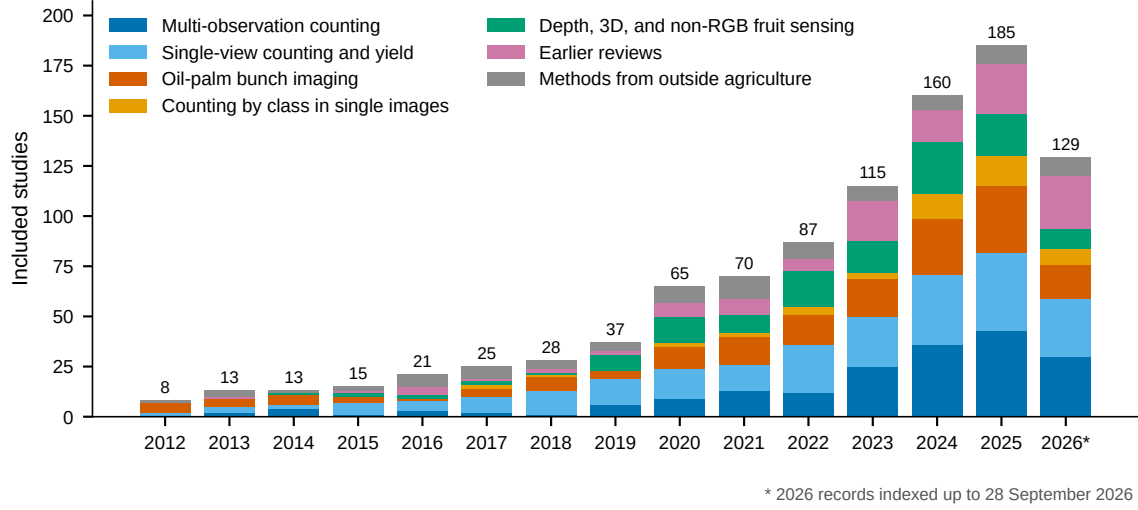


Fig. 2. Included studies by year and group. Multi-observation counting grew from fewer than five studies per year before 2019 to 43 in 2025.

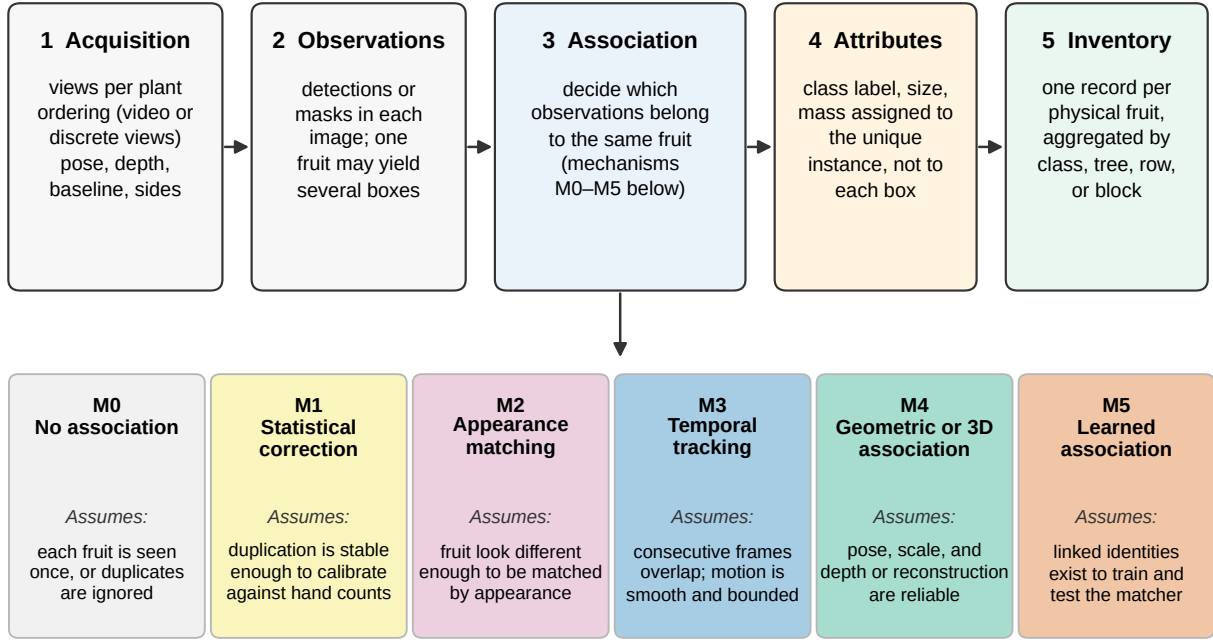


Fig. 3. Five stages from acquisition to inventory, and the six association mechanisms with the condition each assumes.

IV. ACQUISITION: HOW MANY VIEWS, AND FROM WHERE

The first design decision is how the fruit are observed. Eight studies compared acquisition designs on the same plants (Table II); their results show what more views buy and what they cost.

Three results are consistent. First, a single image per side sees a minority of the fruit on a dense canopy: 27% for one side and 54% for two sides in the mango study of Stein *et al.* [1], 40% of harvest with two sides in the study of Wang *et al.* [2], and 58% accuracy for one side in the apple study of Gongal *et al.* [22]. Second, the low recall of single views does not prevent a precise estimate if it is calibrated: in the

study of Stein *et al.*, the dual-view counts correlated with the true counts more tightly ($R^2 = 0.94$) than the multi-view counts ($R^2 = 0.90$), but they had to be scaled up by a factor near two [1]. Precision and accuracy therefore answer different questions; a correlation with hand counts does not show that duplicates were removed. Third, adding views does not always help. In the citrus study of Vijayakumar *et al.* the second side did not reduce the per-tree yield error [26], and in the vineyard study of Ariza-Sentís *et al.* the multiple-view flights recovered more bunches but suffered from motion blur [24]. Zhu *et al.* found that the best number of views per citrus tree (two, four, or six) depended on how heavily the tree had been pruned

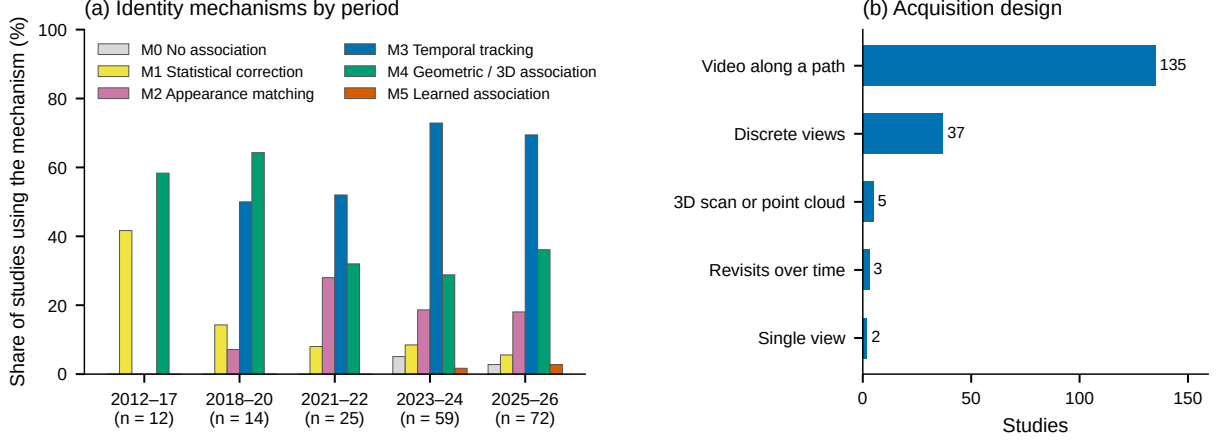


Fig. 4. (a) Share of multi-observation method studies using each mechanism, by period; a study may use several. (b) Acquisition design of the same studies.

TABLE II
STUDIES THAT COMPARED ACQUISITION DESIGNS ON THE SAME PLANTS. VALUES ARE AS REPORTED BY THE AUTHORS.

Study	Crop, scale	Designs compared	Reported result	Reference
Stein <i>et al.</i> [1]	Mango, 16 trees	One image of one side; one image per side; multi-view tracking with triangulation (M4)	Fruit counted: 27% (single, $R^2 = 0.81$), 54% (dual, $R^2 = 0.94$), 101.4% (multi-view, $R^2 = 0.90$)	Hand count on tree and at harvest
Wang <i>et al.</i> [2]	Mango, 21 trees	One image per side; video tracking (M3)	40% of harvest (bias-corrected RMSE 21.7 fruit/tree) vs 62% (RMSE 18.0)	Harvest count
Gongal <i>et al.</i> [22]	Apple, tall spindle	One side; both sides with 3D duplicate removal (M4)	Accuracy 58% vs 82%; error in identifying duplicate apples 21.1%	Manual count
Qureshi <i>et al.</i> [23]	Apple fruitlets	One-sided vs two-sided scans from an over-canopy robot	Counting accuracy 73.7% vs 81.17%	Manual count
Ariza-Sentís <i>et al.</i> [24]	Grape, commercial vineyard	Single-view vs multiple-view UAV video with tracking	Tracked/ground-truth ratio 23% vs 74%; multi-view degraded by motion blur	Annotated video
Felip-Pomés <i>et al.</i> [25]	Apple, 420 trees	Two RGB-D sensors, three distances, two dates, one or both sides	Best MAPE 6.91%, $R^2 = 0.733$ at 225 cm; both sides improved estimates per row stretch	Manual count
Vijayakumar <i>et al.</i> [26]	Citrus, per tree	UAV features plus fruit counts from one or two sides	MAPE 23.45% (one side) vs 25.72% (two sides), both better than UAV only (35.59%)	Harvest per tree
Anderson <i>et al.</i> [27]	Mango, 20 orchards	Raw multi-view counts vs multi-view corrected with hand counts	Average error on packhouse count 11% vs 17%; trellis orchards overestimated by 25.2%	Packhouse count

[28].

Acquisition also decides which association mechanisms are possible. Video along a row gives the frame-to-frame overlap that M3 needs. Discrete views taken around a tree, as in SawitMVC [3] or the eight-sided mango protocol of Nenavath *et al.* [29], have wide baselines and no temporal order, so M3 does not apply and association must rely on appearance, geometry, or learning. Several groups have treated view selection itself as the problem: active view planning for apple clusters [30], viewpoint planning with an octree of fruit regions [31], and graph-based view sequences for sweet pepper [32]. These studies optimize coverage; they do not report counts against a reference.

V. EVIDENCE BY MECHANISM

Table III collects the comparisons of association mechanisms that were made on the same data. The subsections below discuss each mechanism in turn.

A. M0 and M1: counting per image and correcting statistically

The earliest multi-image systems summed counts and calibrated them. Nuske *et al.* registered images from a vehicle driving along vineyard rows and related berry counts to harvest yield; with calibration data from a prior harvest the estimate was within 4%, and within 3% with destructive samples taken at the time of imaging [19]. Across four seasons their estimates captured up to 75% of the spatial variance in yield [46]. Linker photographed 42 apple trees from two sides at three heights, summed the six counts per tree, and calibrated the sum against

TABLE III
COMPARISONS OF ASSOCIATION MECHANISMS ON THE SAME DATA. VALUES ARE AS REPORTED BY THE AUTHORS.

Study	Crop	Compared	Reported result	Metric level
Zhang <i>et al.</i> [33]	Citrus	OrangeSort (motion, counting region) vs SORT vs DeepSORT	Counting MAE 0.081 vs 0.45 vs 1.212	Count
Gené-Molá <i>et al.</i> [34]	Apple	ByteTrack vs SORT vs DeepSORT (YOLOv5x)	HOTA 0.689, 0.680, 0.682; DeepSORT 128 ms per image vs about 15 ms; MAPE 7.47% one week before harvest	Identity and count
Piazolo <i>et al.</i> [35]	Grape (UAV)	SORT vs SORT+ with extended matching cascade	Counting accuracy 33% to 96%; ID switches −62%; MOTA and IDF1 +5–6%	Identity and count
Ponce-Machete <i>et al.</i> [36]	Calamansi (UAV)	SORT, DeepSORT, ByteTrack	ByteTrack best: MOTA 59.67%, HOTA 65.42%, IDF1 73.72%	Identity
Wei <i>et al.</i> [37]	Apple (stereo)	OC-SORT vs ByteTrack with GPS and visual odometry	OC-SORT best (HOTA 0.59); total error 20 apples, RMSE 6.24%, $R^2 = 0.96$	Identity and count
Qi <i>et al.</i> [38]	Tomato clusters	SORT, DeepSORT, ByteTrack, BoT-SORT at reduced frame rates	Public MOT dataset with 165,172 cluster instances	Identity
Liu <i>et al.</i> [39]	Mango, citrus	Tracking alone vs tracking plus SfM correction (M3 vs M3+M4)	SfM step rejected double-counted tracks and improved counting accuracy	Count
Gené-Molá <i>et al.</i> [40]	Apple, 11 trees	2D detection vs detection projected into SfM (M4)	F1 0.816 to 0.881 against 1,455 hand-counted apples	Detection in 3D
Gené-Molá <i>et al.</i> [41]	Apple	LiDAR from one position vs multi-view, with and without forced air	6.5% (multi-view) and 6.7% (air flow) more fruit located	Detection in 3D
Lei <i>et al.</i> [42]	Apple, 60 trees	4D association (position, appearance, topology) vs baselines	Association precision +23.7%; counting accuracy 96.9% for 1,790 apples	Identity and count
Hong <i>et al.</i> [43]	Persimmon	3D Gaussian splatting vs NeRF-based counting	Counting accuracy 96%, reported as 16% above NeRF-based methods	Count
Xing <i>et al.</i> [44]	Cherry tomato	Tracking with and without multi-view compensation	Counting error reduced by 95.6%	Count
Mollineda <i>et al.</i> [45]	Orange	Regression from single frames vs video segments	Multi-frame segments better, tested with nonparametric statistics	Count

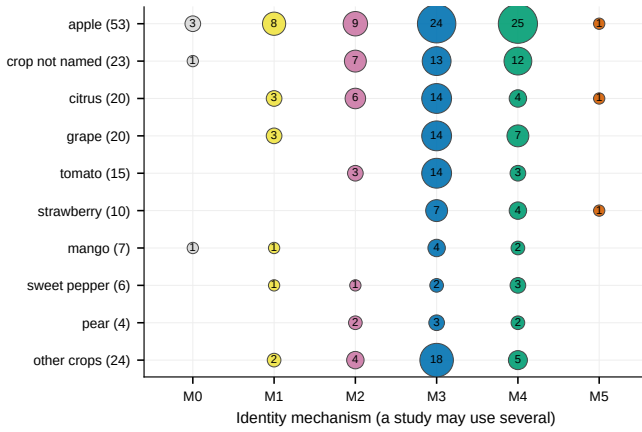


Fig. 5. Multi-observation method studies by crop and mechanism. The number in each circle is the count of studies; the row label gives the number of studies for the crop.

14 trees; the orchard estimate was within 10% of the actual yield although single-tree errors were sometimes large [20]. Song *et al.* aggregated repeated, incomplete observations of pepper fruit in over 28,000 images of more than 1,000 plants with a statistical clustering step and reached a correlation of 74.2% with manual counts without linear adjustment [47].

The strength of M1 is repeatability. Stein *et al.* found the dual-view counts more tightly correlated with true counts than any other design, and Anderson *et al.* reported absolute errors

of 2% or less in repeated night-time imaging of mango orchards [1], [27]. The weakness is transfer. The factor describes an average tree, cannot say which fruit was counted twice, and must be refitted when canopy, cultivar, or acquisition changes; Anderson *et al.* found the error on trellised orchards (25.2%) to be several times that on hedge or single-leader systems (3.4–8.2%), and correcting multi-view counts with hand counts of a stratified sample increased the average error from 11% to 17% [27]. Regression from partial views to total counts remains common in UAV work: Apolo-Apolo *et al.* related top-view apple counts to total counts per tree with $R^2 = 0.80$ [48], and Li *et al.* used a regression across three views to account for blueberries seen more than once [49].

B. M3: temporal tracking

Tracking is now the default. Most studies follow tracking by detection: a detector finds fruit in each frame, a Kalman filter predicts where each track should appear next, and an assignment step, usually the Hungarian algorithm on overlap or distance, links detections to tracks [21], [50], [51]. A counting rule then decides when a track becomes a fruit. Two rules dominate: count each unique track ID, or count a track when it crosses a line or region. Parico *et al.* found the unique-ID rule more reliable for pears (F1 87.85%), because detection flicker caused the line rule to miss fruit that were detected [52]; Zhang *et al.* counted citrus only when a track entered a fixed region, and this rule gave the lowest error in Table III [33].

The main question inside M3 is whether appearance features help. Two apple and citrus studies that compared trackers on the same data found that DeepSORT, which adds a learned appearance descriptor to the motion model, did not outperform trackers that use motion alone and cost far more computation (Table III) [33], [34]. Saraceni *et al.* designed AgriSORT to use motion only, arguing that fruit are static targets of similar appearance, and released a table-grape benchmark with strong self-similarity [53]. Piazzolo *et al.* improved SORT with a matching cascade over Mahalanobis, IoU, and Euclidean distances and reached accuracy comparable to DeepSORT without identity training data [35]. The same issue appears outside agriculture: DanceTrack was built because targets with uniform appearance and diverse motion defeat trackers that rely on appearance [54]. Studies that reported good results with DeepSORT exist, for example banana bunches with RGB-D and three constraints (96.82% counting success) [55], but they did not compare against a motion-only tracker on the same data.

The failure modes that tracking studies report are specific. Missed detections break tracks and create new IDs; Zhang *et al.* added trajectory prediction to bridge such gaps in blueberry video [56], and Zhou *et al.* added a multiple-mapping step for occluded strawberries [57]. Camera shake on a legged robot reduced tracking accuracy until camera-motion compensation added 2.9 points of MOTA [58], and low frame rates from UAVs motivated motion-direction costs and ORB-based compensation in PineSORT [59]. Wang *et al.* measured the net effect in the most transparent way, by reporting double counts (9.9%) and misses (7.3%) separately instead of only their difference [2]. Tracking cannot, by itself, resolve the third source of double counting: a fruit seen from both sides of a row appears in two separate passes. Wang *et al.* listed it as a separate source of double counts [2], and Liu *et al.* used the depth of each fruit relative to its tree trunk to drop fruit that belong to the far side [18].

C. M2: appearance matching

Appearance matching is the only mechanism in two studies and is combined with others in 30, mostly through DeepSORT. Xia *et al.* matched patches of fruit between sequence images with a one-to-one assignment and reached F1 0.816 for apples and 0.864 for oranges in identifying repeated fruit [60]. Kirk *et al.* used re-identification to count soft fruit through occlusions [61]. Zheng *et al.* combined image appearance and position in a learned distance and clustered strawberry detections from overlapping UAV images; the result found fruit that an orthomosaic hid [62]. The pattern across these studies is that appearance works when combined with position and fails as the only cue for look-alike fruit, which agrees with the tracker comparisons above and with the person re-identification literature, where discriminative identity features are learned from large sets of labelled identities [63], [64].

D. M4: geometric and 3D association

Geometric association is the route with the longest history of calibration-free counts. Wang *et al.* used a stereo rig at night

on an autonomous vehicle and registered sequential images; block-level errors were -3.2% for about 480 red-apple trees and 1.2% for about 670 green-apple trees [65]. Stein *et al.* established pairwise correspondences between images from the trajectory of a navigation system, triangulated tracked fruit, and used LiDAR to assign each fruit to its tree; 522 trees and 71,609 mangoes were scanned and the error for individual trees was 1.36% [1]. Liu *et al.* replaced the navigation and LiDAR suite with a monocular camera: fruit and trunks were tracked in 2D, converted into 3D landmarks by semantic structure from motion, and projected back into every image to detect double tracks [18], [39]. Roy *et al.* built a piecewise incremental SfM for apple rows because standard SfM failed under the constrained geometry of a row [66], then merged counts from both sides of 252 trees with an accuracy of $91.98\text{--}94.81\%$ [67]. Dong *et al.* addressed the same two-side problem by aligning the reconstructions of both sides with global and semantic features, since the two sides share little overlap [68]. Gongal *et al.* used 3D positions to remove apples seen from both sides, but misidentified 21.1% of duplicates [22].

Newer M4 systems move the whole count into 3D. Gené-Molá *et al.* projected Mask R-CNN detections into an SfM point cloud and improved F1 from 0.816 to 0.881, at a processing cost they judged unsuitable for real time [40]. Matos *et al.* combined SfM with bipartite graph matching for apples filmed with unconstrained camera motion and low frame rates [69]. Lee *et al.* fused YOLO detections with SLAM point clouds from LiDAR in a 6 m by 70 m pear plot and reached 96.2% counting accuracy [70]. Nellithimaru and Kantor built object-level SLAM from stereo for grape counting (F1 0.977 against image counts) [71]. Rapado-Rincón *et al.* tracked partial point clouds of tomatoes in 3D and estimated the total count with a maximum error of 5.08% [72]; they also showed that the final count of a tracker can hide false positives and false negatives that offset each other, and they introduced tracking metrics to expose them. Implicit representations are the latest variant: FruitNeRF trains a semantic neural radiance field from posed images and clusters the extracted fruit points [73]. FruitNeRF++ adds instance embeddings to remove fruit-specific tuning [74], and Gaussian splatting was reported to count persimmons with 96% accuracy, 16% above NeRF-based counting [43]. These methods build on general reconstruction tools [75], [76], [77]; their evaluations are still dominated by synthetic scenes and a few real trees.

Geometry fails in known places. Scale is undefined for monocular reconstruction unless another sensor fixes it. Pose drift enlarges the region within which two observations are declared the same fruit, and dense clusters then merge. Foliage moves between views, which violates the static-scene assumption of standard SfM and SLAM; dynamic-scene methods exist [78], [79], but no included fruit study measured how much foliage motion degrades counts. Assigning fruit to trees is a separate geometric problem; Zine-El-Abidine *et al.* solved it by aligning a harvest-time point cloud to a leaf-off winter cloud in which each tree's branches are visible (accuracy above 95%) [80].

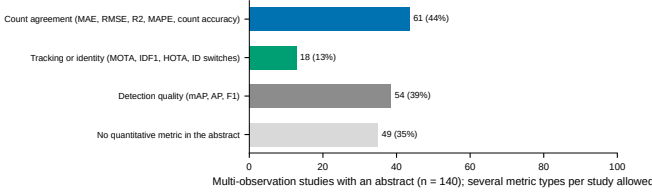


Fig. 6. Metric types reported in the abstracts of multi-observation method studies.

E. M5: learned association

Three studies learn the association. Fusaro *et al.* segmented fruit in coloured terrestrial laser scans, extracted a descriptor per fruit with a sparse convolutional network, and matched fruit between sessions with an attention-based network and a probabilistic assignment, on strawberry and apple data [81]. Isobe *et al.* fused features from different angles of a 360-degree mandarin video in a transformer and regressed the count, without an explicit identity for each fruit [82]. Zheng *et al.*, discussed under M2, learned a distance that combines appearance and position [62]. Outside agriculture, learned association is established for multi-camera tracking of people and vehicles, where datasets with identities across cameras exist [83], [84], [85], [86]. The obstacle in fruit is the same one these datasets removed: M5 needs identity labels across views, and in the included corpus only a handful of fruit datasets provide them, among them GrapeMOTS [87], the tomato MOT set of Qi *et al.* [38], the 4D orchard set of Lei *et al.* [42], and SawitMVC [3].

F. How the evidence was measured

Of the 140 multi-observation method studies with an abstract, 61 (44%) report agreement between estimated and reference counts, 54 (39%) report detection quality, 18 (13%) report an identity metric such as MOTA, IDF1, HOTA, or ID switches, and 49 (35%) report no quantitative metric in the abstract (Figure 6). Identity metrics were developed for exactly this purpose: HOTA balances detection and association accuracy [88], and IDF1 was designed for identities that persist across cameras [83]. A count error alone cannot separate a correct count from offsetting duplicates and misses [72], [2]. The reference also varies: hand counts on the tree measure what the camera could have seen, harvest and packhouse counts include fruit hidden from every view, and annotated images measure only what an annotator saw. Results from different references are not comparable, which is why Tables II and III report the reference or metric level for every row.

VI. CLASS ATTRIBUTES ON UNIQUE FRUIT

A class-wise inventory needs a class for each unique fruit. Two bodies of evidence bear on this. The first is the 49 studies that detect and count fruit by class in single images. Most concern tomato (18) and blueberry (7), and 43 of the 49 use maturity as the class. MacEachern *et al.* detected green, red, and blue wild blueberries with YOLOv4 and predicted yield by regression from the counts [89]; Ni *et al.* counted

blueberries with $R^2 = 0.886$ against ground truth [90]; Deng *et al.* released 17,955 annotated ripe and unripe blueberries in canopy images [91]; and Shimomoto *et al.* predicted sweet pepper yield from average counts at two ripeness stages [92]. These studies count classes in the image, not classes of unique fruit.

The second is the 19 multi-observation studies that report counts per class. Most attach the class during tracking. Ogundele *et al.* used a class-aware ByteTrack for strawberries; counts correlated with ground truth at $R^2 = 0.914$ overall and at $R^2 = 0.950$ for ripe fruit [93]. Zhou *et al.* reported a relative counting error of 6.7%, with 8.7% for ripe and 9.9% for unripe strawberries [57]. Wu *et al.* reached an average counting accuracy of 72.49% across blue, purple, and green blueberries in video [94]. Viveros Escamilla *et al.* counted sweet peppers of four colours with 85.7% accuracy in two simulated environments [95]. Other studies classify after tracking: Hassan *et al.* tracked oranges with ByteTrack and classified each tracked fruit as ripe or unripe with a separate network [96], and Pardo-Beainy *et al.* assigned each tracked strawberry to one of nine size–maturity categories using metric dimensions from RGB-D and GNSS [97].

Two properties of class-wise counting follow from these results. First, class errors and identity errors interact. A fruit seen in ten frames can receive different labels in different frames, especially near class boundaries, and a class-aware tracker that splits the track when the label changes creates a duplicate. None of the 19 abstracts describes how a label is chosen when frames disagree, and none reports a class confusion matrix computed on unique fruit. Second, per-class errors differ from the overall error; the ripe class was counted more accurately than the others in two strawberry studies [93], [57], and in oil palm a video-based detector reached 90.56% mAP for a single class but 70.21% for several ripeness classes [98]. An inventory that reports only a total hides both effects.

VII. DEPTH AND OTHER MODALITIES

The 129 depth and 3D sensing studies use RGB-D cameras (74), stereo (19), LiDAR (13), or other non-RGB signals. Their main use is not association but detection, localization, and sizing of individual fruit, most often apples (46). Depth was used to remove background: Fu *et al.* filtered objects farther than 1.2 m from a Kinect V2 in a dense fruiting-wall orchard before detection [99]. It was fused with colour inside the detector: Tu *et al.* trained separate colour and depth detectors for passion fruit and fused them late, raising F1 from 0.885 for a standard Faster R-CNN on RGB-D images to 0.946 [100], and Kaukab *et al.* filled depth holes before fusion because outdoor depth images are of low quality [101]. It gave metric size: Wang *et al.* estimated on-tree mango dimensions from RGB-D with RMSEs of about 4–5 mm but noted that the camera could not be used in direct intense sunshine [102], and Neupane *et al.* compared eight depth cameras against a 20 mm bias-corrected RMSE requirement for fruit localization and sizing [103]. LiDAR detected apples by reflectance with F1 0.858 [104], and LiDAR–camera fusion reduced localization error to one fifth of that of a RealSense D455 [105]. Active

thermal imaging with water mist separated immature citrus from leaves of similar colour and supported tracking-based counting in video [106].

For association, depth enters through M4. Zheng *et al.* filtered detections by depth and fused robot odometry before inter-frame prediction for tomatoes, reaching MOTA 0.954 against YOLOv8 with DeepSORT [107]. Abeyrathna *et al.* obtained 3D coordinates of each tracked apple from a RealSense D455 at three vehicle speeds and three camera angles [108]. Felip-Pomés *et al.* found that the Azure Kinect gave more consistent results than the ZED 2 across dates in a 420-tree orchard [25]. Liu *et al.* used depth relative to the trunk to resolve two-side duplicates [18]. Monocular depth networks now offer depth without a depth sensor [109], [110], and one included study used such a network with amodal masks to measure apple diameter [111], but none used predicted depth for cross-view association with a counted reference. The evidence therefore supports depth as an aid to separating neighbouring fruit and to placing fruit in a common frame when its error is small relative to fruit spacing, and it identifies the conditions (sunlight, range, registration) under which that error grows.

VIII. OIL PALM

The 170 oil-palm studies concentrate on ripeness (Figure 7). Seventy-six classify the ripeness of images of single bunches, 37 detect bunches in a scene, 21 study non-RGB sensing of ripeness, 12 address quality, defects, mass, or pose, 10 detect loose fruit, 9 count bunches, and 4 are dataset papers. The setting is on the palm in 40 studies, at a mill, conveyor, or laboratory in 25, on the ground in 13, and from a UAV in 4; 88 abstracts do not state it. Class labels are attached to instances in 132 studies. Only 11 use more than one view of the same bunch or tree.

Ripeness classification has a long record. Early work extracted colour features and classified them with neural networks or support vector machines [112], [113], followed by colour and texture features with classical classifiers [114] and by convolutional networks deployed on mobile devices [115]. The largest study in the corpus trained on 10,000 photographs from five Indonesian plantation sites in four classes and reported 96.54% accuracy on locations not seen in training [116]. Reviews of this work agree that computer vision with deep learning is the most feasible field method [10] and that fusion with LiDAR may add localization [11]. Non-RGB sensing has been tested with thermal imaging [117], hyperspectral imaging [118], laser-light backscattering [119], LiDAR [120], and radar with bunch mass [121], several of them on harvested bunches.

Detection on the palm moved from single-class to multi-class ripeness detectors. Lai *et al.* trained YOLOv4 on images from a RealSense D435 with expert labels for ripe and unripe bunches [122], Mansour *et al.* compared detectors on four ripeness classes [123], Junos *et al.* detected bunches, grabbers, and palms in plantation scenes and bunches in UAV images [124], [125], and Zaman *et al.* addressed occlusion by fronds in UAV images [126]. Goh *et al.* released 400 plantation

images with registered depth maps and point clouds from a RealSense D455f for ripeness classification and localization [127], the only oil-palm RGB-D dataset in the corpus. At the mill, Suharjito *et al.* graded piles of bunches from smartphone video [128] and released annotated datasets of piles [129].

Counting is the gap. The nine counting studies count bunches in single images of palms [130], [131], [132], [133], [134], count moving bunches on a sorting line with a detector and tracker [135], count unstripped bunches in a mill with Faster R-CNN and a Euclidean-distance tracker [136], count bunches and loose fruit together for harvest management [137], or describe an IoT-based counting system [138]. Naftali *et al.* and Hamdani *et al.* built plantation datasets with partial visibility, occlusion, and density levels [130], [131], and Hidayat *et al.* proposed a standard procedure for capturing bunch images in the plantation [132]. None of the counting studies associates bunches across images of the same palm. Suharjito *et al.* recorded unharvested palms from multiple angles with a smartphone for a digital census and released 14,503 images in five maturity classes, but the annotations are per image [139]. The only oil-palm dataset in the corpus with cross-view identity links is SawitMVC [3]. Warni *et al.* combined stereo detection with Hungarian matching to localize loose fruit in 3D [140], which is an association step, but for a stereo pair, not for views around a palm. Tree-level counting from UAV images has addressed re-counting errors for palms themselves [141], [142], not for bunches.

The oil-palm case differs from the orchard cases in ways that matter for mechanism choice. Bunches of the same class on one palm look alike, which weakens M2. A census photographs a palm from a few discrete positions, which removes the temporal order that M3 needs. Fronds occlude bunches differently from each side, and the canopy is tall, which makes depth from consumer RGB-D cameras less reliable at the required range. The census classes are defined by months to harvest, so neighbouring classes differ in subtle colour changes, and a class error on one view can split one bunch into two records of different classes.

IX. POSITION RELATIVE TO EARLIER REVIEWS

Table IV lists the earlier reviews that are closest to this one. The column “Stated method” records what each abstract states about how studies were found. The last column records whether identity across observations is the organizing variable; it is not, in any of them, although several discuss occlusion and extrapolation from partial counts. Reviews of tracking and re-identification outside agriculture [143], [144], [64] and of 3D reconstruction [79], [145] provide the mechanisms but not the agricultural evidence.

X. GAPS AND A MEASUREMENT AGENDA

The synthesis supports five gaps, each tied to a measurement.

G1: Identity is rarely measured. Only 13% of the multi-observation studies report an identity metric in the abstract. A study that claims to remove duplicates should report, on annotated sequences or view sets, the numbers of duplicate

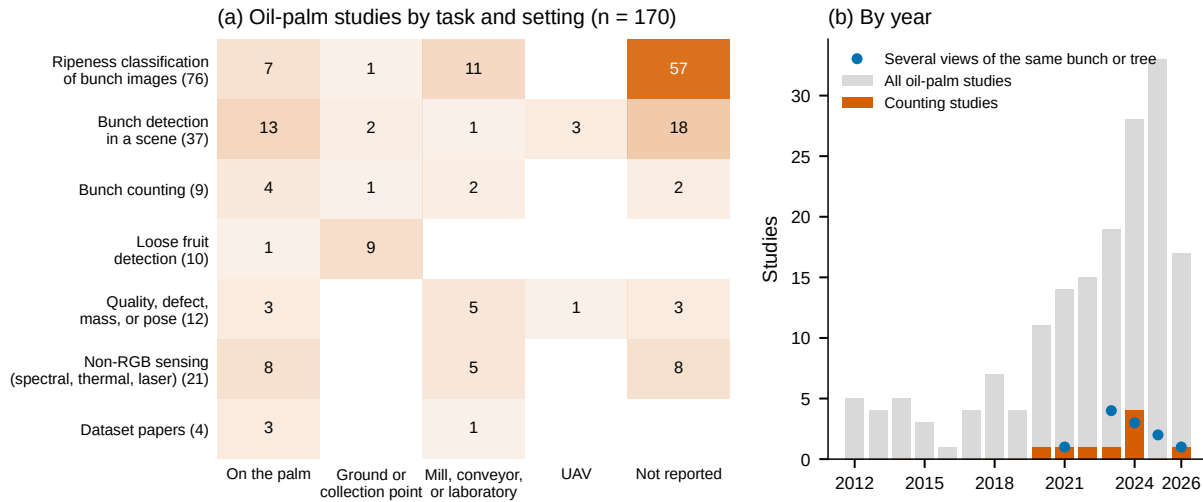


Fig. 7. Oil-palm studies. (a) Task by setting. (b) Studies per year, with counting studies and studies that use several views of the same bunch or tree.

TABLE IV
EARLIER REVIEWS CLOSEST TO THIS REVIEW.

Review	Scope	Organizing variable	Stated method	Identity across views as organizing variable
Gongal <i>et al.</i> 2015 [4]	Fruit detection and localization for harvesting and crop-load estimation in apples, pears, citrus	Sensor and system	Not stated	No
Koirala <i>et al.</i> 2019 [6]	Deep learning for fruit detection and yield estimation; extrapolation of image counts for occluded fruit	Method and application	Not stated	No
Fu <i>et al.</i> 2020 [5]	Consumer RGB-D cameras for fruit detection and localization	Depth principle and sensor	Not stated	No
Anderson <i>et al.</i> 2021 [8]	Forecasting fruit load and harvest timing	Estimation route (sampling, vision, remote sensing, models)	Not stated	No
Maheswari <i>et al.</i> 2021 [146]	Semantic segmentation for fruit yield estimation	Network architecture	Not stated	No
Khan <i>et al.</i> 2021 [147]	Machine learning in oil palm, 2011–2020	Task (regression, classification)	Systematic, 61 papers	No
Lai <i>et al.</i> 2023 [10]	Oil-palm FFB ripeness detection	Sensing modality and classifier	Systematic, 51 papers	No
Goh <i>et al.</i> 2025 [11]	FFB ripeness classification	Sensing technique	Not stated	No
Pu <i>et al.</i> 2025 [148]	UAV counting of field crops	Agronomic target and model type	52 key studies	No
Rong <i>et al.</i> 2026 [149]	Orchard yield estimation with multi-source sensing	Data source	PRISMA	No
This review	Turning repeated fruit observations into unique, class-labelled records; oil palm as the main case	Identity mechanism and its assumptions	Systematic map, Scopus, 971 studies	Yes

records, missed fruit, false merges of two fruit into one, and ID switches, in addition to the count error. Wang *et al.* [2] and Rapado-Rincón *et al.* [72] show that this is feasible.

G2: Mechanisms are rarely compared on the same data. Table III contains 13 such comparisons, most within M3. Comparisons across mechanisms, for example tracking against 3D association on the same trees, are rare, and comparisons against a calibrated M1 baseline are rarer, although calibration against hand counts is the usual way of turning image counts into orchard estimates [6], [8]. A new mechanism should be tested against M1 fitted on separate trees.

G3: Class-wise identity is almost absent. Nineteen multi-observation studies report counts per class, and none reports

how label disagreement between views is resolved. The required measurement is a per-class count error computed on unique fruit, together with a confusion matrix of classes on unique fruit.

G4: Discrete-view protocols lack mechanisms. Most methods assume video. Census protocols photograph a plant from a few fixed positions, as in oil palm [3] or in multi-sided mango and citrus imaging [29], [28]. For such data, M3 does not apply, and the evidence on M2, M4, and M5 comes mostly from other acquisition designs. Wide-baseline association with known or estimated pose is the missing experiment.

G5: Oil palm has detection and ripeness evidence but no identity evidence. A bunch-level census from images needs

an evaluation on palms photographed from several sides, with identity links, class labels in census classes, and a per-class count reference. One dataset now provides this [3].

These gaps suggest a minimum reporting set for future work: acquisition protocol (views per plant, ordering, pose source, depth source, baseline); mechanism and the component that removes duplicates; reference type (hand count on the tree, harvest, or annotated images); count error per plant and per class; identity metrics on annotated data; and an M1 baseline fitted on separate plants.

XI. LIMITATIONS OF THIS REVIEW

Scopus was the only source; work indexed only in Web of Science, IEEE Xplore, or preprint servers is missing unless Scopus also indexes it. Abstracts were not available from the database export used and were retrieved from other bibliographic services; 254 records without an abstract were decided from titles, sources, and machine-generated summaries. Screening and coding were done by one reviewer working with a large language model, without a second independent reviewer; the screening decisions are provided as supplementary material so that they can be audited, and an inter-rater check on a random sample is the first step needed to strengthen the map. Rule-based codes for crop, modality, platform, and metric types were spot-checked but not verified for every record, and metric types were read only from abstracts, so a study may report a metric in its full text that is not counted here. Open-access full text was available for about a third of the included studies (331 of 971), so most coding relied on abstracts. The set of methods papers from outside agriculture is purposive, limited to the most cited records of each run. Results were not pooled, and the numbers in Tables II and III should be compared only within a row.

XII. CONCLUSION

Counting fruit from images is an identity problem as much as a detection problem. The 187 multi-observation studies in this map use six mechanisms to decide which observations belong to the same fruit. Statistical correction is repeatable but must be refitted and cannot name duplicates. Temporal tracking dominates recent work, and on look-alike fruit, motion-only trackers were not outperformed by trackers with appearance features. Geometric association gives calibration-free counts when pose, scale, and depth are reliable, at the cost of sensors or reconstruction time. Learned association is new and limited by the lack of identity-labelled data. The evidence is measured mostly by count agreement and detection quality; identity metrics and per-class counts on unique fruit are rare. In oil palm, the literature covers ripeness classification and detection, but no study associates bunches across views of a palm. A study that wants to show a duplicate-aware, class-wise inventory should state its acquisition assumptions, test its mechanism against a calibrated baseline on separate plants, and report identity errors and per-class counts on unique fruit.

DATA AVAILABILITY

The search strings are listed in Appendix B. The screening decision for every record, the coded evidence matrix, and the analysis scripts that reproduce the numbers and figures are provided as supplementary material.

DECLARATION OF GENERATIVE AI USE

A large language model (Claude, Anthropic) read titles and abstracts during screening, recorded screening decisions and codes, wrote the analysis scripts, and drafted the text under the direction of the authors, who take responsibility for the content.

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APPENDIX A
EVIDENCE MATRIX FOR MULTI-OBSERVATION STUDIES

Table V lists every multi-observation study included in the review. The result reported by each study is given in the supplementary evidence matrix.

TABLE V: Evidence matrix for the 187 multi-observation studies. Acq.: V video along a path, D discrete views, S 3D scan, T revisits over time, I single view. Platform: GV ground vehicle or robot, UAV, HH handheld or smartphone, LB laboratory or conveyor, FX fixed camera (– not stated in the abstract). Mechanisms M0–M5 as in Section III. Class: counts reported per class. Metrics: types reported in the abstract (count agreement, tracking or identity, detection quality); n.a., no abstract available.

Study	Year	Crop	Platform	Acq.	Mechanism	Class	Metrics
Qian [150]	2013	apple	–	D	M1	–	n.a.
Wang [65]	2013	apple	GV	V	M4	–	none
Gongal [151]	2014	apple	–	D	M4	–	n.a.
Nuske [46]	2014	grape	GV	V	M1	–	none
Nuske [19]	2014	grape	GV	V	M1	–	none
Song [47]	2014	pepper	–	D	M1	–	none
Linker [20]	2015	apple	–	D	M1	–	none
Gongal [22]	2016	apple	–	D	M4	–	none
Roy [66]	2016	apple	–	D	M4	–	none
Stein [1]	2016	mango	–	V	M4	–	count, det
Roy [30]	2017	apple	GV	D	M4	–	none
Roy [152]	2017	apple	–	D	M4	–	n.a.
Liu [39]	2018	–	–	V	M3+M4	–	count
Huang [153]	2019	–	GV	V	M2+M3	–	none
Jarvinen [154]	2019	apple	–	V	M3	–	count
Liu [18]	2019	mango	–	V	M3+M4	–	det
Nellithimaru [71]	2019	apple	–	V	M4	–	det
Roy [67]	2019	apple	–	V	M4+M1	–	none
Wang [2]	2019	mango	–	V	M3	–	count
Apolo-Apolo [48]	2020	apple	UAV, GV	D	M1	–	count
Dong [68]	2020	–	–	D	M4	–	none
Gené-Mola [41]	2020	apple	–	S	M4	–	count
Gené-Mola [40]	2020	apple	–	D	M4	–	det
Gené-Mola [155]	2020	apple	–	D	dataset	–	none
Gené-Mola [156]	2020	apple	–	S	dataset	–	none
Santos [157]	2020	grape	–	V	M3+M4	–	det
Serafino [158]	2020	citrus	GV	V	M3	–	none
Sun [159]	2020	cotton	–	D	M4	–	count
Anderson [27]	2021	mango	–	D	M1	–	count
Gao [160]	2021	apple	–	V	M3	–	n.a.
Gao [161]	2021	apple	–	V	M3	–	det
Itakura [162]	2021	apple	–	V	M3	–	det
Kirk [61]	2021	–	–	V	M2	–	n.a.
Osman [163]	2021	apple	GV	V	M2+M3	–	det
Osman [164]	2021	apple	–	V	M2+M3	–	det
Parico [52]	2021	pear	–	V	M2+M3	–	det
Scalisi [165]	2021	apple	–	V	M4	–	n.a.
Torres-sánchez [166]	2021	grape	UAV, GV	S	M4	–	count
Zaenker [31]	2021	pepper	GV	D	M4	–	none
Zhang [167]	2021	pomegranate	GV	I	M4	–	none
Zine-El-Abidine [80]	2021	apple	–	S	M4	–	none
Dai [168]	2022	tomato	GV	V	M3	–	n.a.
Gao [169]	2022	apple	–	V	M3	–	n.a.
Ge [170]	2022	tomato	GV	V	M2+M3	yes	det
He [171]	2022	–	–	V	M3	–	n.a.
Marangoz [172]	2022	pepper	GV	V	M4	–	det
Turečková [173]	2022	tomato	–	V	M3	–	n.a.
Vulpi [174]	2022	–	GV	D	M4	–	n.a.
Wang [175]	2022	apple	UAV	V	M3	–	none
Xia [60]	2022	apple	–	V	M2	–	count, det
Zhang [33]	2022	citrus	–	V	M3+M2	–	count, det
Zhang [176]	2022	apple	–	D	M1	yes	count
Zhu [177]	2022	citrus	–	D	M4	–	n.a.
Abeyrathna [108]	2023	apple	GV	V	M2+M3+M4	–	count, det
Ariza-Sentís [178]	2023	grape	UAV, GV	V	dataset	–	none
Ariza-Sentís [179]	2023	grape	UAV, GV	V	M3	–	none
Arlotta [180]	2023	grape	GV, LB	V	M3+M4	–	none
Ciarfuglia [181]	2023	grape	GV	V	M3+M4	–	none
Gené-Mola [34]	2023	apple	–	V	M3+M2	–	count, track, det
Gremes [182]	2023	citrus	GV	V	M3	–	count, det
Guo [183]	2023	kiwifruit	–	V	M3	–	n.a.

Study	Year	Crop	Platform	Acq.	Mechanism	Class	Metrics
Hu [184]	2023	apple	–	V	M3	–	count, det
Lawal [185]	2023	strawberry	–	V	M3	–	det
Li [49]	2023	blueberry	GV	D	M1	–	count
Lin [186]	2023	citrus	–	V	M2+M3	–	count
Lyu [187]	2023	grape	–	V	M3	–	n.a.
Parico [188]	2023	pear	–	V	M2+M3	–	n.a.
Qureshi [23]	2023	apple	GV	D	M4	–	count, det
Rapado-Rincón [72]	2023	tomato	GV	V	M4+M3	–	none
Shen [189]	2023	grape	–	V	M3	–	n.a.
Vijayakumar [26]	2023	citrus	UAV	D	M1	–	count
Villacrés [190]	2023	apple	–	V	M3	–	n.a.
Wu [191]	2023	apple	–	V	M3	–	count, track, det
Xie [192]	2023	–	GV	D	M4	–	none
Zaenker [32]	2023	pepper	GV	D	M4	–	none
Zheng [62]	2023	strawberry	–	D	M5+M4	–	none
Zheng [193]	2023	citrus	UAV, GV	V	M2+M3	–	count, det
Zhou [194]	2023	strawberry	–	V	M3	–	count
Ahmad [195]	2024	apple	UAV, GV, HH	V	M0	–	none
Ahmedt-Aristizabal [196]	2024	grape	–	V	M3+M1	–	none
An [197]	2024	strawberry	–	V	M3	–	n.a.
Ariza-Sentís [87]	2024	grape	UAV	V	dataset	–	none
Ashokan [198]	2024	–	–	V	M2+M3	–	none
Du [199]	2024	pepper	–	V	M2+M3	–	count, track, det
Feng [200]	2024	citrus	–	V	M3	–	count
Garderes [201]	2024	apple	–	V	M3+M1	–	track, det
Hernandez [202]	2024	–	–	V	M3	–	n.a.
Ishizuka [203]	2024	–	–	V	M3+M4	–	none
Kang [204]	2024	tomato	GV, HH	V	M3	–	count, det
Magalhães [205]	2024	tomato	LB	V	M4	–	count
Matos [69]	2024	apple	GV	V	M4+M3	–	none
Meyer [73]	2024	apple	–	D	M4	–	none
Muzaddid [206]	2024	cotton	UAV, GV	V	M3	–	none
Nenavath [29]	2024	mango	–	D	M0	–	count
Pai [207]	2024	tomato	–	V	M3	yes	none
Qi [208]	2024	tomato	–	V	M3	yes	n.a.
Safre [209]	2024	cherry	–	V	M2+M3	–	n.a.
Santos [210]	2024	citrus	–	V	M3+M4	–	n.a.
Saraceni [53]	2024	grape	GV	V	M3	–	none
Scalisi [211]	2024	pear	–	V	M4	yes	n.a.
Sekharamantray [212]	2024	apple	UAV	V	M3	–	count, det
Tu [213]	2024	passion f.	–	V	M2+M3	–	n.a.
Villacrés [214]	2024	apple	GV	D	M0	–	n.a.
Viveros [95]	2024	citrus	–	V	M2+M3	yes	none
Wang [215]	2024	pitaya	–	V	M4	–	n.a.
Wang [216]	2024	camellia	–	V	M3	–	n.a.
Xia [217]	2024	kiwifruit	–	V	M3	–	n.a.
Yang [218]	2024	apple	–	V	M2+M3	–	count, track, det
Yuan [219]	2024	strawberry	–	V	M4	–	count, det
Zhang [58]	2024	peach	GV	V	M3	–	count, track
Zheng [220]	2024	citrus	–	V	M3	–	n.a.
Zhou [221]	2024	citrus	–	D	M4	–	count
Zhou [57]	2024	strawberry	–	V	M3	yes	count
Zhu [28]	2024	citrus	–	D	M1	–	none
Akella [222]	2025	coconut	–	V	M3	yes	det
Ariza-Sentís [24]	2025	grape	UAV	D	M3	–	count
Cao [223]	2025	apple	UAV	V	M2+M3	–	count, track, det
Dong [224]	2025	camellia	–	V	M3+M1	–	det
Garg [225]	2025	–	UAV, GV	V	M4	–	count
Hassan [96]	2025	citrus	HH	V	M3+M4	yes	none
Hobart [226]	2025	apple	UAV	S	M4	–	none
Hong [43]	2025	persimmon	–	D	M4	–	count
Isobe [82]	2025	citrus	–	V	M5	–	none
Issad [227]	2025	–	–	V	M3	–	det
Jiang [228]	2025	cotton	GV, HH	D	M4	–	count, det
Kemper [229]	2025	grape	–	V	M3	–	count
Kutyrev [230]	2025	apple	UAV, GV	D	M1	–	det
Lei [42]	2025	apple	–	T	M4+M2	–	count
Louis [231]	2025	melon	GV, LB	V	M3	–	count
Lu [232]	2025	litchi	–	V	M2+M3	–	count
Lyu [233]	2025	grape	–	V	M3	–	n.a.
Lyu [234]	2025	grape	–	V	M3	–	n.a.
Majdalawieh [235]	2025	–	–	V	M2+M3	–	det
Matos [236]	2025	apple	–	V	M3	–	n.a.
Mendez [237]	2025	pepper	GV	V	M3	–	track, det
Meyer [74]	2025	apple	–	D	M4	–	none

Study	Year	Crop	Platform	Acq.	Mechanism	Class	Metrics
Mollineda [45]	2025	citrus	GV	V	M1	–	none
Ogundele [93]	2025	strawberry	–	V	M3	yes	count, det
Pichhika [238]	2025	mango	–	V	M3	–	n.a.
Ponce-Machete [36]	2025	–	UAV	V	M3+M2	–	track, det
Qi [38]	2025	tomato	–	V	M3+M2	–	count, track, det
Sasse [239]	2025	grape	HH	V	M3	–	det
Tu [240]	2025	passion f.	–	V	M3	–	n.a.
Vaishnavi [241]	2025	citrus	–	V	M2+M3	yes	count
Wang [242]	2025	tomato	–	V	M3	yes	count
Wei [37]	2025	apple	–	V	M3+M4	–	count, track
Wu [94]	2025	strawberry	–	V	M3	yes	count, det
Xie-Li [59]	2025	pineapple	UAV	V	M3	–	track, det
Xing [243]	2025	tomato	GV	V	M2+M3	yes	track, det
Yan [244]	2025	apple	–	V	M2+M3	–	det
Yang [245]	2025	–	UAV, GV	V	M0	–	none
Yang [246]	2025	–	UAV, GV, LB	V	M4	–	none
Zhai [247]	2025	–	–	V	M2+M3	–	count, track
Zhang [248]	2025	kiwifruit	HH	V	M3	–	n.a.
Zhang [56]	2025	blueberry	–	V	M3	–	det
Zheng [107]	2025	tomato	GV	V	M3+M4	yes	track
Zhou [55]	2025	banana	–	V	M2+M3	–	none
Bharti [249]	2026	apple	–	V	M3+M4	–	det
Blondet [250]	2026	avocado	–	V	M3	–	n.a.
Chen [251]	2026	apple	–	I	M0	–	count, det
Farhoud [252]	2026	citrus	–	V	M3	yes	count, det
Felip-Pomés [25]	2026	apple	–	V	M3+M4	–	count, det
Feng [253]	2026	tomato	–	V	M3	–	n.a.
Fusaro [81]	2026	apple	GV	T	M5	–	none
Gao [254]	2026	–	GV	V	M3+M4	–	none
Hu [255]	2026	strawberry	–	D	M4	–	n.a.
Huang [256]	2026	mango	–	V	M3	yes	n.a.
Indriani [3]	2026	oil palm	HH	D	dataset	yes	none
Lee [70]	2026	pear	–	V	M3+M4	–	count, det
Lin [257]	2026	tomato	–	V	M3	yes	none
Nguyen [258]	2026	apple	UAV, GV, LB	S	M4+M1	–	count
Pardo-Beainy [97]	2026	strawberry	–	V	M3+M4	yes	count
Park [259]	2026	–	–	V	M4	–	n.a.
Piazolo [35]	2026	grape	UAV, GV	V	M3	–	count, track
Santos [260]	2026	–	GV	V	M3+M4	–	none
Si [261]	2026	citrus	HH	V	M2+M3	–	count, track, det
Tu [262]	2026	passion f.	–	V	M3	–	n.a.
Wang [263]	2026	apple	UAV, LB	V	M3+M4	–	none
Wang [264]	2026	–	UAV, GV	D	M4	–	count
Weng [265]	2026	citrus	UAV	V	M3	–	n.a.
Woo [266]	2026	grape	HH	V	M4	–	det
Xing [44]	2026	tomato	GV	V	M3	–	count
Yang [267]	2026	grape	–	V	M4	–	count, det
Yoshida [268]	2026	grape	GV	T	M4	–	none
Zhao [269]	2026	–	–	D	M4	–	none
Zhao [270]	2026	apple	GV	D	M4	–	det
Zhu [271]	2026	–	GV	V	M2+M3	–	track

APPENDIX B

SCOPUS SEARCH STRINGS

Each string was submitted to the Scopus advanced search on 28 September 2026 (UTC). Runs Q8a–Q8e and Q9 were sorted by citation count.

Q1 (871 records): TITLE-ABS-KEY((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND (count* OR "yield estimat*" OR "yield predict*" OR "yield forecast*" OR "load estimat*" OR "fruit load" OR "crop load" OR inventor* OR enumerat* OR "fruit number") AND ("multi-view" OR multiview OR "multiple view*" OR "multi-camera" OR "multiple camera*" OR video* OR "image sequence*" OR track* OR "structure from motion" OR "structure-from-motion" OR SfM OR SLAM OR "3D reconstruct*" OR "point cloud*" OR "double count*" OR duplicat* OR "re-identification" OR "data association" OR "cross-view" OR "both sides" OR "two sides") AND (image* OR camera* OR vision OR "deep learning" OR "neural network*" OR detect* OR segment* OR YOLO OR "R-CNN" OR photograph*)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DECI) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")

Q2 (757 records): TITLE((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR

kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND (count* OR "yield estimat*" OR "yield predict*" OR "load estimat*" OR "fruit load" OR "crop load" OR "fruit number")) AND TITLE-ABS-KEY((image* OR camera* OR vision OR "deep learning" OR "neural network*" OR detect* OR segment* OR YOLO OR "R-CNN" OR photograph*)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DEC) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")

Q3 (1,079 records): TITLE-ABS-KEY(("oil palm" OR "Elaeis guineensis" OR "fresh fruit bunch*" OR "palm oil fruit*") AND (image* OR vision OR camera* OR "deep learning" OR "machine learning" OR detect* OR convolutional OR YOLO OR segment* OR classif* OR ripeness OR maturity OR count*) AND (bunch* OR fruit* OR ripeness OR maturity OR harvest*)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DEC) OR SUBJAREA(BIOC))

Q4 (361 records): TITLE-ABS-KEY((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND (ripeness OR maturity OR "ripening stage*" OR "maturity stage*" OR "size class*" OR grade* OR grading OR defect* OR "multi-class" OR "per class" OR "class-wise") AND (count* OR "yield estimat*" OR inventor* OR "load estimat*") AND (detect* OR track* OR segment*)) AND (image* OR camera* OR video* OR vision)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DEC) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")

Q5 (1,988 records): TITLE-ABS-KEY((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND ("RGB-D" OR RGBD OR "depth camera*" OR "depth image*" OR "depth map*" OR "depth sensor*" OR "depth information" OR "stereo camera*" OR "stereo vision" OR binocular OR LiDAR OR "time-of-flight" OR "point cloud*" OR Kinect OR RealSense OR "3D camera*") AND (detect* OR count* OR locali* OR "yield estimat*" OR "size estimat*") AND (orchard* OR field* OR tree* OR canopy OR plant* OR greenhouse* OR vineyard*)) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DEC) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")

Q6 (236 records): TITLE-ABS-KEY(("multi-object tracking" OR "multiple object tracking" OR "multi-target tracking" OR "multiple target tracking" OR "re-identification" OR "multi-camera tracking" OR "multi-camera multi-object" OR "multi-view detection" OR "multiview detection" OR "cross-view association" OR "multi-view association" OR "cross-camera association" OR "object counting" OR "structure from motion" OR "visual SLAM" OR "instance association") AND (review OR survey OR benchmark)) AND DOCTYPE(re) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI))

Q7 (1,049 records): TITLE-ABS-KEY((fruit* OR orchard* OR "tree crop*" OR "fresh fruit bunch*" OR "oil palm" OR "Elaeis guineensis" OR apple* OR citrus OR orange* OR mango* OR grape* OR vineyard* OR strawberr* OR tomato* OR kiwifruit* OR pear OR pears OR peach* OR cherr* OR blueberr* OR "sweet pepper*" OR capsicum OR banana* OR pineapple* OR "date palm*" OR coconut* OR pomegranate* OR avocado* OR lemon* OR plum OR plums OR almond* OR walnut* OR olive*) AND (count* OR "yield estimat*" OR "yield predict*" OR "load estimat*" OR detect* OR ripeness OR maturity OR harvest* OR locali*) AND (image* OR vision OR camera* OR sensor* OR "deep learning" OR "machine learning")) AND DOCTYPE(re) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(AGRI) OR SUBJAREA(COMP) OR SUBJAREA(ENGI) OR SUBJAREA(MULT) OR SUBJAREA(PHYS) OR SUBJAREA(EART) OR SUBJAREA(ENVI) OR SUBJAREA(MATH) OR SUBJAREA(DEC) OR SUBJAREA(BIOC)) AND NOT TITLE-ABS-KEY("fruit fl*" OR drosophila OR "blood count*")

Q8a (3,604 records; the 25 most cited were retained): TITLE("multi-object tracking" OR "multiple object tracking" OR "multi-target tracking" OR "tracking-by-detection" OR "tracking by detection") AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q8b (2,243 records; the 25 most cited were retained): TITLE("multi-view" OR multiview OR "multi-camera" OR "multiple cameras" OR "cross-view") AND TITLE(detection OR tracking OR association OR counting OR "re-identification") AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q8c (9,696 records; the 25 most cited were retained): TITLE("structure from motion" OR "structure-from-motion" OR SLAM OR "bundle adjustment" OR "multi-view stereo") AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q8d (1,554 records; the 25 most cited were retained): TITLE("object counting" OR "crowd counting" OR "counting objects" OR "class-agnostic counting") AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q8e (6,435 records; the 25 most cited were retained): TITLE("re-identification" OR "reidentification") AND TITLE(object OR vehicle OR person OR instance) AND PUBYEAR > 2011 AND PUBYEAR < 2027 AND (SUBJAREA(COMP) OR SUBJAREA(ENGI)) AND (DOCTYPE(ar) OR DOCTYPE(cp))

Q9 (1,082 records; the 25 most cited were retained): (TITLE("simple online and realtime tracking") OR TITLE("deep association metric") OR TITLE("BoT-SORT") OR TITLE("StrongSORT") OR TITLE("multiview detection with feature perspective transformation") OR TITLE("representing scenes as neural radiance fields") OR TITLE("3D Gaussian splatting for real-time radiance field rendering") OR TITLE("depth anything") OR TITLE("segment anything") OR TITLE("vision transformers for dense prediction") OR TITLE("end-to-end object detection with transformers") OR TITLE("DETRs beat YOLOs")) AND PUBYEAR > 2011 AND PUBYEAR < 2027